

DIHAS SATHNINDU

Software Engineer · Backend · Distributed Systems · Full-Stack

Colombo, Sri Lanka +94 76 424 4692 dihas.sathnindu@gmail.com linkedin.com/in/dihas-sathnindu
github.com/dihass dihas.site

SUMMARY

Computer Science graduate (First Class Honours) and former IFS Software Engineering Intern, focused on building scalable, server-authoritative backends and distributed real-time systems. Strong in Java, C#, and Python with hands-on experience in concurrency, REST API design, automated testing, and the full SDLC across enterprise and production projects.

EDUCATION

BEng (Hons) Computer Science — First Class Honours

IIT / University of Westminster · Colombo, Sri Lanka

Sep 2022 – Jun 2026

- Relevant coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Software Engineering, Object-Oriented Programming, Machine Learning.
- Edexcel A-Levels, Leeds International School: Mathematics (A), Physics (B), Chemistry (C).

TECHNICAL SKILLS

Languages	Java, C#, Python, TypeScript, JavaScript, PL/SQL, SQL, Swift
CS Foundations	Data Structures, Algorithms, System Design, Design Patterns, OOP, Concurrency, SDLC
Backend	.NET, Node.js, Express, SignalR, REST APIs, real-time and server-authoritative systems
Frontend	React.js, Next.js, Tailwind CSS, SwiftUI
Data & Storage	PostgreSQL, MongoDB, Oracle Database, Redis, query optimisation, indexing, caching
Testing	Unit, Integration, and System Testing, TAR Automation, test case design
DevOps / VCS	Git, GitHub Actions, CI/CD Pipelines, Docker, Vercel, Netlify
Practices	Agile/Scrum, Jira, code review, technical documentation, software security

EXPERIENCE

Software Engineering Intern | IFS · Colombo, Sri Lanka

Sep 2024 – Sep 2025

- Built and shipped features across enterprise ERP modules using Java and the Marble framework on an Oracle PL/SQL backend, contributing through the full SDLC from technical design to release.
- Diagnosed and resolved complex production defects through systematic debugging and root-cause analysis, strengthening stability across core enterprise modules.
- Automated unit, integration, and system tests with the TAR framework, expanding test coverage and reducing manual QA effort.
- Raised code quality through design and code reviews, applying architectural patterns and design principles within a distributed global Agile team.

Founder & Software Engineer | DarwinCode · Colombo, Sri Lanka

Jun 2026 – Present

- Founded a software studio building full-stack web platforms for hospitality clients, leading every stage from requirements and architecture to deployment and client delivery.
- Shipped production websites with version-controlled CI/CD workflows, performance and SEO optimisation, and a focus on secure, maintainable code.

PROJECTS

Omi Game Engine — Real-Time Multiplayer Game Backend · .NET 10 · SignalR · Redis · Distributed Systems

2026

- Architected a server-authoritative multiplayer backend on .NET 10 with SignalR WebSocket gameplay, Redis-backed lobby and session state, and JWT authentication, enforcing all game logic server-side to prevent client tampering.
- Designed for concurrency and consistency using distributed per-lobby locking, private per-player hand delivery, and seamless reconnect handling, backed by a fully tested Omi rules engine (trump selection, follow-suit validation, trick resolution, scoring, dealer rotation).
- Containerised the backend with Docker and deployed a Vite/React frontend on Vercel, hardening production with CORS, health checks, rate limiting, and secure state synchronisation.

Stack: .NET 10, C#, SignalR, Redis, JWT, React, Vite, Docker **Live:** <https://omicardgame.vercel.app/>

Leakage-Free Multimodal Sepsis Prediction · *Python · PyTorch · Deep Learning · FYP* 2025 - 2026

- Built a multimodal deep-learning pipeline predicting sepsis onset 6 hours in advance, reaching AUROC 0.9283 across 26,607 ICU patients (MIMIC-IV) by fusing a two-layer LSTM with Bio_ClinicalBERT embeddings through a meta-learner.
- Engineered a strict leakage-free data pipeline optimised for memory footprint and large-scale clinical data, with SHAP explainability, full technical documentation, and a Streamlit demo.

Stack: Python, PyTorch, LSTM, Bio_ClinicalBERT, SHAP, Streamlit, MIMIC-IV **GitHub:** <https://github.com/dihass/LF-MSP>

PropVRty — Full-Stack Real Estate Marketplace · *Node.js · MongoDB · React · Full-Stack* 2024

- Built a full-stack marketplace with a Node.js/Express REST API, MongoDB, and a React frontend, integrating Panolens.js 360-degree virtual tours and the Uploadcare media API.
- Applied REST design, database schema modelling, and OOP across the full SDLC, deploying to production on Vercel.

Stack: MongoDB, Express, React.js, Node.js, JavaScript, REST APIs, HTML5, CSS3 **Live:** <https://prop-vr-ty.vercel.app/>

LocateLink — Real-Time Location Sharing · *Next.js · TypeScript · Redis · Backend* 2026

- Engineered a real-time location-sharing service on Next.js/TypeScript with Upstash Redis for low-latency session state, UUID-based secure link generation, and Leaflet.js map rendering.

Stack: Next.js, TypeScript, Upstash Redis, REST APIs, Leaflet.js, UUID

WeatherDashboard — Native iOS Weather Explorer · *Swift · SwiftUI · Mobile* 2025

- Built a native SwiftUI iOS app on the OpenWeather REST API with async/await networking, CoreLocation geocoding, MapKit, and SwiftData caching for offline access.

Stack: Swift, SwiftUI, MapKit, CoreLocation, SwiftData, OpenWeather API **GitHub:** <https://github.com/dihass/WeatherDashboard>

REFERENCES

Ashfaq Zeemath

Senior Lead Software Engineer · Iconique (Pvt) Ltd
ashfaqz@iconique.tech +94 75 464 0451

Saumya Alexandra

Senior Software Engineer · IFS
saumya.alexandra@ifs.com +94 72 595 5845